

Manual Execution for Problems 3 and 4

Graph with 3 connected components

Source file: 'data/manual/undirected_components_1.txt'

Vertices: '0 1 2 3 4 5 6 7'

Edges:

- 0 -- 1

- 1 -- 2

- 2 -- 3

- 3 -- 0

- 0 -- 2

- 4 -- 5

- 5 -- 6

- 4 -- 6

Vertex '7' is isolated.

Problem 3: DFS connected components

Traversal:

Step	Stack before pop	Popped	Visited after step
1	0	0	'0'
2	3, 2, 1	1	'0, 1'
3	3, 2, 3, 2	2	'0, 1, 2'
4	3, 2, 3, 3	3	'0, 1, 2, 3'

First component: '{0, 1, 2, 3}'

Restart from the smallest unvisited vertex:

Step	Stack before pop	Popped	Visited after step
5	4	4	'0, 1, 2, 3, 4'
6	6, 5	5	'0, 1, 2, 3, 4, 5'
7	6, 6	6	'0, 1, 2, 3, 4, 5, 6'

Second component: '{4, 5, 6}'

Restart again:

Step	Stack before pop	Popped	Visited after step
8	7	7	'0, 1, 2, 3, 4, 5, 6, 7'

Third component: '{7}'

Problem 4: BFS connected components

Traversal:

Step	Queue before pop	Popped	Visited after step
1	0	0	'0, 1, 2, 3'
2	1, 2, 3	1	'0, 1, 2, 3'
3	2, 3	2	'0, 1, 2, 3'
4	3	3	'0, 1, 2, 3'

First component: '{0, 1, 2, 3}'

Restart from '4':

Step	Queue before pop	Popped	Visited after step
5	4	4	'0, 1, 2, 3, 4, 5, 6'
6	5, 6	5	'0, 1, 2, 3, 4, 5, 6'
7	6	6	'0, 1, 2, 3, 4, 5, 6'

Second component: '{4, 5, 6}'

Restart from '7':

Step	Queue before pop	Popped	Visited after step
8	7	7	'0, 1, 2, 3, 4, 5, 6, 7'

Third component: '{7}'

Graph with 1 connected component

Source file: 'data/manual/undirected_components_2.txt'

Edges:

- 0 -- 1
- 1 -- 2
- 2 -- 3
- 3 -- 4
- 4 -- 5
- 5 -- 0
- 0 -- 2
- 1 -- 4

Problem 3: DFS connected components

Starting from '0', the traversal reaches every vertex:

'0 -> 1 -> 2 -> 3 -> 4 -> 5'

Connected components: '{0, 1, 2, 3, 4, 5}'

Problem 4: BFS connected components

Starting from '0', the queue expands in this order:

'0 -> 1 -> 2 -> 5 -> 3 -> 4'

Connected components: '{0, 1, 2, 3, 4, 5}'